Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/17/2013
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Integrated Regulation of Bone and Muscle Metabolism by Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23284-3067	**Congressional District:**	3
Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lang, Charles (The Pennsylvania State University)		
Grant/Contract No.:	NCC 9-58-MA02802		
Performance Goal No.:			
Performance Goal Text:			
	Original project aims/objectives Our long-term goal is to understand the structural and functional relationship between muscle atrophy (i.e., sarcopenia) and bone loss (i.e., osteopenia) which occurs as a result of extended exposure to the spaceflight environment. The specific objectives of this proposal are to: 1) Examine the time course for structural, functional, and molecular changes occurring in muscle and bone in response to mechanical unloading; 2) Examine the effect of inhibiting, or exaggerating, unloading-induced muscle loss on unloading-induced bone loss; 3) Examine the effect of inhibiting, or exaggerating, unloading-induced bone loss on unloading-induced muscle loss; and 4) Evaluate potential therapeutic strategies for prevention of muscle and bone loss during unloading. Upon completion of our specific aims, and based on our experimental approaches, we will recommend specific therapeutic strategies and countermeasures that target muscle and bone as an integrated unit and respect their inherent interdependency. To accomplish these objectives, we will use hindlimb suspension (HLS), a well established in vivo model of simulated microgravity unloading. Our overall		

Task Description:	hypothesis is that unloading-induced osteopenia is at least partially dependent on unloading-induced sarcopenia and, conversely, unloading induced sarcopenia is at least partially dependent on unloading-induced osteopenia. The rationale for this proposed research is that there has not been a comprehensive and simultaneous examination of the effects of unloading on both muscle and bone. Understanding how these two systems respond to simulated microgravity will allow for the development of effective countermeasures to preserve astronaut health during missions and upon return to Earth. The research team assembled has extensive experience examining, both in vitro and in vivo, bone (Donahue) and muscle (Lang) response to loading and unloading. This collective expertise will bring a unique and valuable perspective to the project. Key findings The first year of the project has proceeded very well with no unanticipated problems. We have completed aim 1 and these findings have been submitted for publication to the Journal of Bone and Mineral Research. The manuscript is now undergoing revision. In completing the first aim we subjected 4-month-old C57BL/6J mice to hindlimb suspension (HLS) for three weeks, sacrificing 12-16 mice on day (D) 0, 7, 14, and 21. Lean mass was 7-9% lower for HLS vs. control from D7-21. Absolute mass of the gastrocnemius (gastroc) decreased 8% by D7, and was maximally decreased 16% by D14 of HLS. Quadriceps (quad) lost a similar amount by D14. Control muscle mass did not change with age. mRNA levels of Atrogin-1 in the gastroc and quad were increased 99% and 122%, respectively, at D7 of HLS. Similar increases in MuRF1 mRNA levels were noted at D7. Both atrogenes returned to baseline by D14. Protein synthesis in gastroc and quad was reduced 30% from D7-14 of HLS, returning to baseline by D21. Cortical thickness and area fraction of the femur and tibia did not decrease until D14 of HLS. Cortical bone of controls did not change with age. HLS mice had lower distal femur bone volume fraction (-22%) and trabecular thickness (-15%) by D14; however, the effects of HLS were often eliminated by D21 due to the decline of trabecular bone mass of controls. Femur strength was decreased approximately 13% by D14 of HLS, with no change in tibia mechanical properties at any time point. Impact T his investigation reveals that muscle atrophy precedes bone loss during unloading and may contribute to subsequent skeletal deficits. Countermeasures that preserve muscle may help to prevent or delay bone loss induced by mechanical unloading. Age-related trabecular bone loss, similar to that which occurs in mature astronauts, is superimposed on unloading. Preservation of muscle mass, cortical structure, and bone strength with age suggests muscle may have a greater effect on cortical than trabecular bone. Proposed research plan for the coming year: During the coming year we will complete aim 2 and begin aim 3.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have demonstrated that sarcopenia precedes osteopenia during unloading. These data suggest that muscle atrophy may contribute to subsequent skeletal deterioration. In addition, the natural decline of trabecular bone with age in normally loaded control mice tends to reduce the magnitude of the effect of HLS, with many trabecular microstructural parameters in unloaded mice converging with controls by day 21. Mounting evidence suggests a dominant role for muscle over bone in the homeostatic maintenance of these two tissues. Muscle size is strongly correlated with bone size and strength in human. Indeed, extension of our findings suggests that a therapeutic strategy that targets muscle may also preserve bone during unloading. Taken together, this study has refined the appropriate use of the HLS model, while providing new insights into the development of effective countermeasures to the musculoskeletal effects of spaceflight, prolonged bed rest, and disuse.
Task Progress:	We have completed aim 1 as listed in the original application. We have begun breeding mice for aim 2 and will begin aim 2 in September of 2013.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Lloyd SA, Loiselle AE, Zhang Y, Donahue HJ. "Connexin 43 deficiency desensitizes bone to the effects of mechanical unloading through modulation of both arms of bone remodeling." Bone. 2013 Jul 24;57(1):76-83. http://dx.doi.org/10.1016/j.bone.2013.07.022 ; PubMed PMID: 23891909 , Jul-2013
Articles in Peer-reviewed Journals	Lloyd SA, Zhang Y, Paul EM, Laufenberg LLJ, Lang CH, Donahue HJ. "Interdependence of muscle atrophy and bone loss induced by mechanical unloading." Journal of Bone and Mineral Research. In revision as of May 2013. , May-2013